

COVID-19 Death Rates by Demographics

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Abstract

In our analysis, we examined our dataset, “Monthly COVID-19 Death Rates per 100,000 Population by Age Group, Race and Ethnicity, Sex, and Region with Double Stratification” to determine whether COVID-19 death rates changed between demographics. Previous studies found that COVID-19 mostly affects the elderly and medically vulnerable. Public health measures are generally effective at reducing the spread of COVID-19. Death rates are mostly uncorrelated with socio-economic class, but racial/ethnic minorities bear a disproportionate burden from COVID-19. To analyze our dataset, we ran a series of one-way ANOVA tests to determine if crude death rates differed significantly across sex, race, and age group. A log regression model was constructed to predict the outcomes of no deaths (0) and at least one death (1). We found that, consistent with previous studies, age was the strongest predictor of death. Sex differences were minimal. Race/ethnicity analyses showed differences in population-normalized crude death rates. Log regression reflected these patterns for Age and Sex, but were limited for Race due to differences in subgroup size and the nature of transforming the data into a binary outcome.

Introduction

The dataset contains 94,248 observations of COVID-19 deaths in regions across the globe, stratified by age group, race/ethnicity, sex and region. Other variables included are data

collection start and end date, crude death rate, 95% confidence intervals, and a subset of the data is double stratified by race or age indicated by a second subgroup. After cleaning, observations with empty or unreliable values (death counts that greater than 0 but less than 20) were removed, resulting in a final data set of 65,439 rows. The original dataset was primarily structured for stratified analyses to explore differences in COVID-19 death rates across the different demographic groups. The motivation for this project is to understand disparities in COVID-19 mortality across demographic groups. Specifically, we aim to answer the research question: Do age, sex, and race significantly affect the likelihood of COVID-19 death? By building statistical models to explore these relationships we can identify high risk populations and provide evidence based guidance for policy makers. This analysis helps to highlight patterns in COVID-19 mortality.

Literature Review

In a study by Ayouni et. al. (2021), scientists systematically reviewed and assessed methods for controlling COVID-19. The goal of the review was to evaluate public health and non-pharmaceutical control measures implemented worldwide to mitigate and control the spread of COVID-19.

PubMed, Science Direct, and MedRxiv were systematically searched for articles related to Covid-19 health measures. Articles that were clinical trials, quasi-experimental, case-control, cohort, longitudinal, and interrupted time series published up to March 16, 2021 were included in the review. Relevant data was collected from each study, including authors, article title, country of origin, aim of study, study design, characteristics of studied population, public health

interventions implemented (description of interventions and controls if applicable), outcomes, and conclusions. Studies were analyzed for bias using the EPHPP quality assessment tool.

18 articles were included in the systematic review, and it was found that public health interventions and non-pharmaceutical control measures were effective in reducing the transmission of COVID-19 and were associated with reduced epidemic growth. The identified studies included in the systematic review showed travel restrictions, border measures, quarantining travelers from affected countries, city lockdown, restrictions of mass gathering, isolation and quarantine of confirmed cases and close contacts, social distancing measures, compulsory mask wearing, contact tracing and testing, school closures, and PPE for health workers were effective in mitigating the spread of COVID-19 to varying degrees.

In the absence of effective treatment and vaccine against COVID-19, these public health measures and non-pharmaceutical interventions reduce infection and mortality, but some are not efficient enough to be implemented alone and, depending on the phase of the epidemic in the country, multiple interventions may be needed to control the outbreak.

Paul et. al. (2021) examines the impact of socio-economic factors on the spread of COVID-19. This study focuses on broad spectrum socio-economic data collected from the 5 years American Census Survey 2018 of the United States Census Bureau. COVID-19 prevalence and death rate data is then correlated with the data from the US Census Bureau to find asymmetries in urban and rural areas, and counties where a large fraction of the population is non-white.

The socio-economic metrics measured include population density, fraction of non-white population, income per capita, fraction of population below poverty line, unemployment rate, fraction of population that does not have health insurance, fraction of population employed,

fraction working in construction, service, delivery, or production, fraction that uses public transit or carpool, mean commute distance, and fraction above 65 years of age. COVID-19 prevalence was measured by confirmed case rate, death rate, and fatality rate.

The study finds that COVID-19 is highly dependent on population density in urban areas, and less so in suburban and rural areas. Regions with larger fractions of racial and ethnic minorities are affected most, particularly in southern regions. COVID-19 affects elderly populations more, but spreads more easily in areas with younger populations. These results are descriptive and not predictive.

Hooper et. al. (2020) is a viewpoint article that proposes the pandemic has exposed health inequities and created an opportunity to address these underlying causes. Geographic locations that report data by race/ethnicity indicate African American and Latino individuals have disproportionate COVID-19 outcomes. The most pervasive disparities were seen in African American, Latino, American Indian, Alaska Native, and Pacific Islander populations. For example, in Chicago, Illinois, mortality rates were substantially higher in African Americans (73 per 100,000) compared to Latino (36 per 100,000) and white residents (22 per 100,000).

These disparities may be caused by underlying comorbidities disproportionately burdened by minority populations such as diabetes, cardiovascular disease, asthma, HIV, morbid obesity, liver disease, and kidney disease, but not for chronic lower respiratory disease or COPD. An additional reason may be that public policies exacerbate health disparities. According to the article, health interventions that are adapted for local contexts and community characteristics are more effective than standard approaches. Since COVID-19 triggered uniform public health recommendations related to social distancing or sheltering-in-place, local contexts and population characteristics may have not been considered, resulting in less effective results.

Hooper et. al. (2020) calls for further studies to improve our understanding of demographic and social determinants of health, pulling representative epidemiological data from ongoing or planned studies using weighted random sampling, standardized racial/ethnic categories, and widespread and accessible testing. Additionally, studies are needed to assess the influence of state and local mitigation policies on health service disparity, the role of community-level protective factors and interventions in mitigating adverse consequences of sector disruptions caused by COVID-19, and the influence of COVID-19-related racism and other types of discrimination.

Data Analysis

Exploratory Data Analysis

Box plots of crude death rates for each factor highlighted the distribution and presence of outliers (See figures [A1-3](#)). Almost all subgroups were highly zero-inflated with many observations falling outside of the 3rd quartile range. Median values reflected this with most subgroups having a median value of zero, particularly among younger age groups and certain racial/ethnic groups.

Histograms of the crude death rates further exemplified these findings (See figures [B1-2](#)). A log transformation was applied, but it did very little to help normalize the distribution. Due to this, results from ANOVA should be taken with a degree of skepticism, as the normality assumption is not validated. Although outside the scope of this course, a Kurskal-Wallis Test was done comparing median values between the subgroups as a non-parametric method for the data.

Zero inflation of the crude rate informed the decision to use a binary outcome in logistic regression analyses. These descriptive statistics guided the selection of variables and subgroups for one-way ANOVA, Kruskal-Wallis, and logistic regression.

I. Age

Among age groups, the mean crude COVID-19 death rate ranged from 0.00045 to 45.03589, with a median of 1.04 deaths across all subgroups. Subgroups 5-11 and 12-17 had the lowest mean crude death rate while older subgroups (e.g., 65-74, 75+) had markedly higher rates. Across all subgroups, the standard deviation was 34.16788 indicating substantial variability. Higher age groups saw increased variability compared to lower age groups.

II. Sex

Descriptive statistics by sex revealed that the mean crude death rate was 5.9013 for males and 4.8607 for females, with median values of 2.49 and 2.29 respectively.

III. Race

Crude COVID-19 death rates vary across race and ethnicity. The mean rates ranged from 2.28897 to 9.15361, with the highest observed in Non-Hispanic American Indian or Alaska Native (AI/AN) and the lowest observed in Non-Hispanic Native Hawaiian or Other Pacific Islander (NH/PI). Standard deviations were largest in AI/AN populations, indicating its high mean death rate may be due to low total population.

Model Building

We conducted a series of one-way ANOVA tests to determine whether crude death rates differed across levels of sex, race, and age group. We also performed a logistic regression on a constructed death present variable that reported 0 if no deaths and 1 if any deaths in the group. Prior to running the ANOVAs, we assessed the normality. Normality was evaluated using Q-Q

plots of model residuals shown in figure [C1-3](#). These plots indicated clear deviations from normality for all three ANOVA models.

Because the normality assumption was not met, the ANOVA results should be interpreted with caution. However, due to this, Kruskal-Wallis tests are added to assess any potential findings.

I. Age

The one-way ANOVA table for age indicated that there existed at least one significant difference in the age groups ($p\text{-value} < 2e-16$). I used a Tukey's HSD test and reported the findings using a Compact Letter Display (CLD) where groups that share a letter have no significant differences between them. 75+ years had the highest crude death rate, significantly higher than all other age groups. 65-74 years had the second-highest rate, significantly lower than 75+ but higher than all younger groups. The remaining years had no statistically significant differences among them. This shows a clear age gradient, with COVID-19 death rates increasing sharply in older adults, particularly above 65 years. The Kruskal-Wallis test corroborated these findings but also added its own insights. Younger age groups (0-17) are non-significant from each other. However, from 18 to 29, there was an indicated significant jump in death rates starting in young adults. The overall trends are very similar between the two methods.

crude_rate groups		
75+ years	4.503589e+01	a
65-74 years	1.245377e+01	b
50-64 years	5.530952e+00	c
40-49 years	2.665101e+00	c
30-39 years	1.109320e+00	c
18-29 years	3.064583e-01	c
0-4 years	1.708054e-02	c
12-17 years	4.688995e-03	c
5-11 years	4.535637e-04	c

For the logistic regression (see figure [D1](#)), using 0-4 years as the reference group, the model showed a clear increase in odds of death with age. Younger age groups, such as 5-17 years had low odds relative to the reference whereas older age groups had substantially higher odds. In particular, 50+ years of age had dramatically elevated odds of death per observation. These results align closely with the crude death rates and ANOVA analyses, reflecting the strong age gradient in COVID-19 mortality.

II. Sex

The one-way ANOVA indicated that mean crude death rates did differ significantly at a 0.05 p-value, but not at a 0.1 p-value. A post-hoc Tukey test was conducted to explore the pairwise differences. The difference between male and female was about 1.04 deaths with a confidence interval of (0.173, 1.908). The Tukey HSD revealed that males had slightly higher rates than females. The Kruskal-Wallis test instead revealed that the crude death rates did not differ significantly. This makes more sense and is corroborated by other findings that death rates usually do not differ by sex.

Logistic regression examining the association between sex and the presence of at least one COVID-19 death indicated that males had 1.004 times the odds of experiencing a death compared to females (95% CI = 0.427, 2.359). See figure [D2](#). This suggests that sex was not meaningfully associated with the presence of at least one COVID-19 death, as the odds were nearly identical between males and females.

III. Race/Ethnicity

For ethnic/racial groups, the one-way ANOVA did indicate that the mean crude death rates differed significantly. A post-hoc Tukey test was conducted to explore the pairwise

differences. Another CLD display is given below, where groups sharing the same letter do not differ significantly. AI/AN individuals had the highest crude death rate, significantly higher than every other group. NH/PI had the lowest crude death rate. Whites, Blacks, and Hispanics, and Asians formed an intermediate cluster with overlapping rates. There is a clear descending gradient of COVID-19 death rates across the racial and ethnic groups. Because of the normality assumption being violated, a Kruskal-Wallis test was done as well. The results largely corroborated the Tukey HSD findings, and similar or matching differences were observed.

	crude_rate	groups
Non-Hispanic American Indian or Alaska Native	9.153610	a
Non-Hispanic Black	6.960245	b
Non-Hispanic White	6.015613	bc
Hispanic	4.913570	cd
Non-Hispanic Asian	4.011818	de
Non-Hispanic Native Hawaiian or Other Pacific Islander	2.288973	e

Examining race/ethnicity using the logistic regression model (see figure [D3](#)) indicated that Hispanic individuals had baseline odds of experiencing at least one COVID-19 death of 17.6. The model indicated that Black and White individuals had the highest odds of experiencing a death in the dataset relative to Hispanics. AI/AN individuals and NH/IP individuals had lower odds relative to Hispanics in this model, despite AI/AN having one of the highest crude death rates. These results differ from the ANOVA and crude rate analyses because the logistic regression models row-level odds per observation and does not take into consideration the population size. Many AI/AN groups had zero deaths due to small population size, which lowers the observed odds despite having high crude death rates. Larger subgroups like White and Black individuals have many rows with at least one death, inflating their odds. Because our ANOVA analysis captured the population risk and instead our logistic regression reflects frequency of

deaths, some subgroups appear to have lower odds than expected. This limitation should be considered when interpreting the differences between subgroups.

Results and Conclusions

Our analysis of COVID-19 death rates across multiple populations stratified by age, sex, and race/ethnicity revealed clear patterns in mortality and risk. Age was the strongest predictor of death. The odds of experiencing at least one death per month increased dramatically with the age of the group, and individuals aged 75+ had the highest risk. Sex differences were minimal, with only slightly higher odds observed among males compared to females. Race/ethnicity analyses showed notable differences in population-normalized crude death rates, with AI/AN exhibiting some of the highest crude rates while NH/IP had the lowest. Logistic regression reflected these patterns for Age and Sex but was limited for Race due to differences in subgroup sizes and the nature of transforming the data into a binary outcome.

These findings largely corroborate the existing literature. Older individuals consistently experience higher COVID-19 mortality, and some racial/ethnic minority groups have elevated risk. This is likely due to structural and social determinants of health. The high crude death rate observed among the AI/AN population aligns with prior studies, even though logistic regression underrepresented this group. Limitations of this study include the absence of true population denominator, the structure of the dataset as it was prestratified, and potential underreporting for smaller subgroups. Improvements could be made in many ways. One way would be by including total population counts to model true proportions of death. Also incorporating additional predictors into the model such as socioeconomic status could expose further trends. Multi-level modelling could also be done to account for the region the data was collected in or time-based clustering because as time goes on, COVID-19 death rates decrease. Despite these clear

limitations, the study provides valuable insights into subgroup differences in COVID-19 mortality and highlights populations at higher risk. This could offer guidance for public health interventions in the future.

Works Cited

Ayan Paul *et al* 2021 *J. Phys. Complex.* 2 035017

Ayouni, I., Maatoug, J., Dhouib, W. *et al.* Effective public health measures to mitigate the spread of COVID-19: a systematic review. *BMC Public Health* 21, 1015 (2021).
<https://doi.org/10.1186/s12889-021-11111-1>

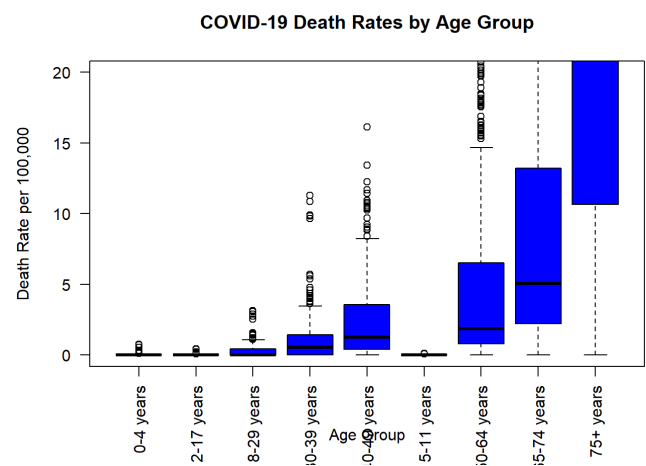
Webb Hooper M, Nápoles AM, Pérez-Stable EJ. COVID-19 and Racial/Ethnic Disparities. *JAMA*. 2020;323(24):2466–2467. doi:10.1001/jama.2020.8598

CDC. (2025, September 17). Monthly COVID-19 Death Rates per 100,000 Population by Age Group, Race and Ethnicity, Sex, and Region with Double Stratification. Retrieved 2025,.

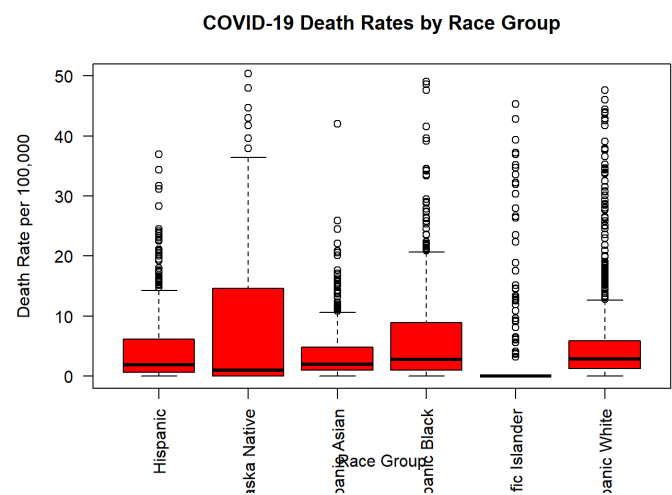
Appendix A: Figures

Figures

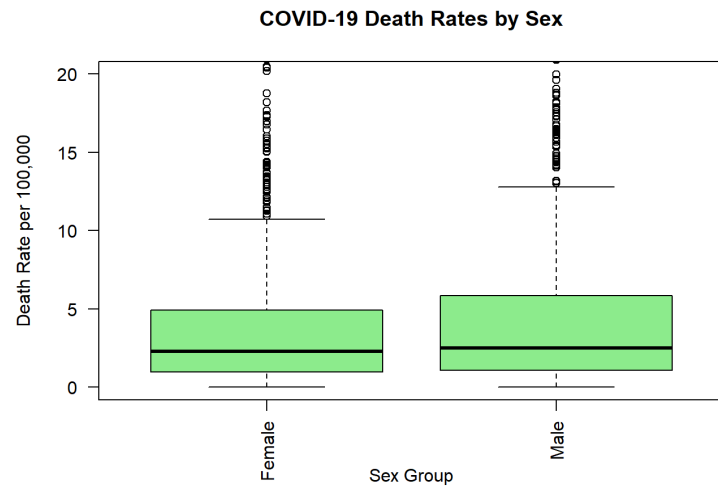
A1: COVID-19 Death Rateby Age



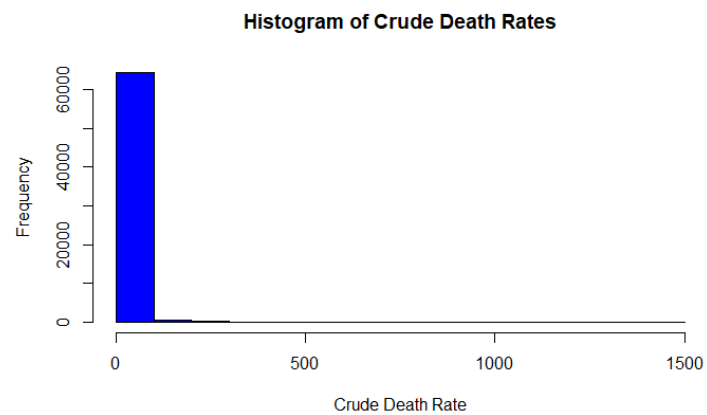
A2: COVID-19 Death Rate by Race



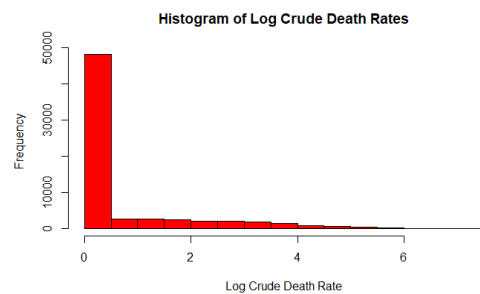
A3: COVID-19 Death Rate by Sex



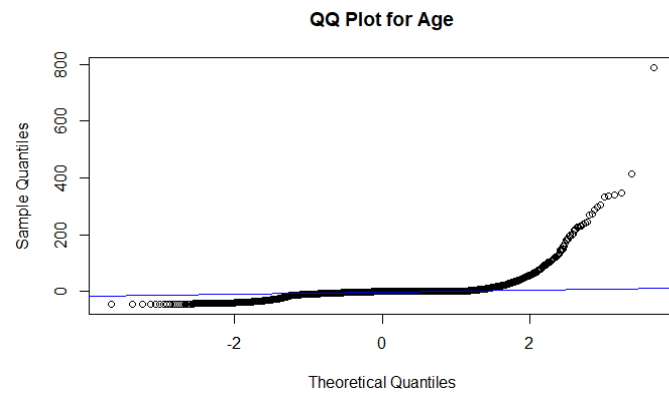
B1: Histogram of Crude Death Rates



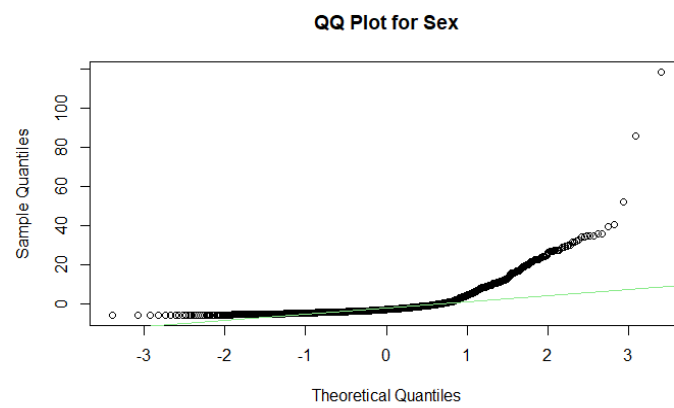
B2: Histogram of Log Crude Death Rates



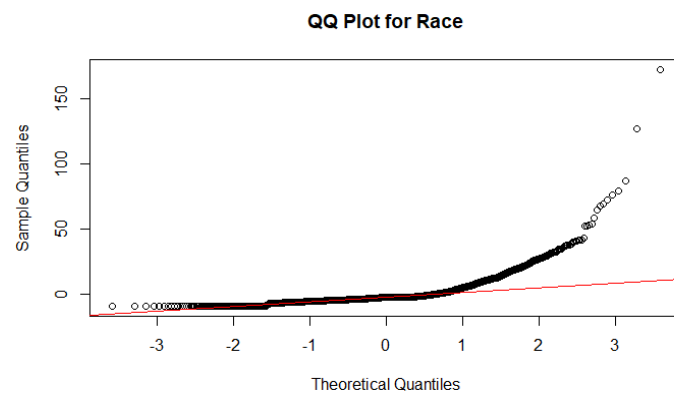
C1: QQ Plot of Age Model



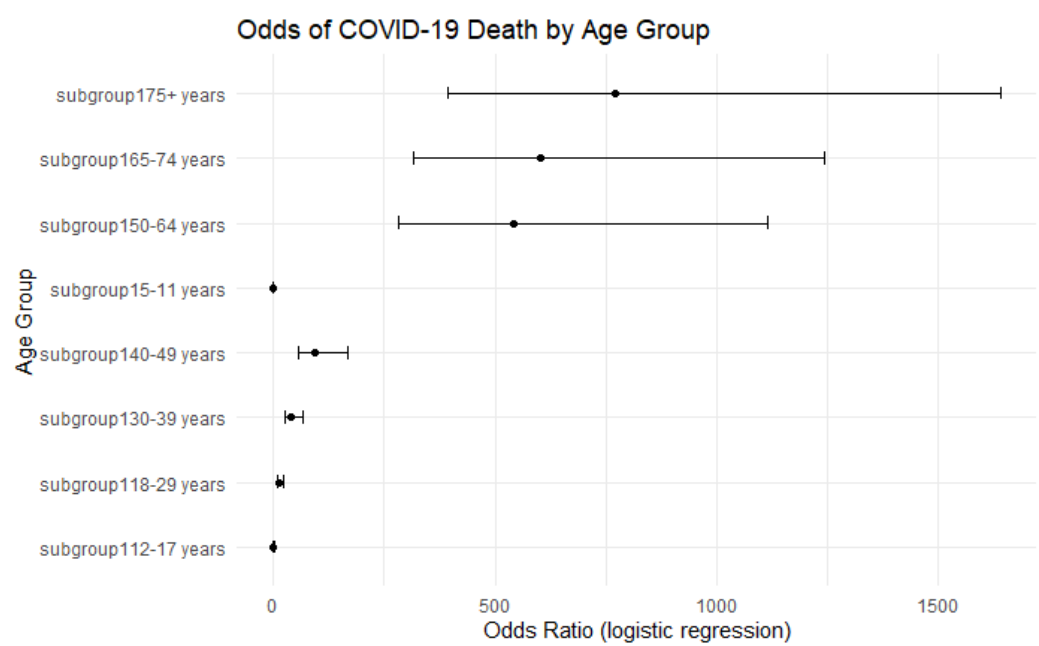
C2: QQ Plot of Sex Model



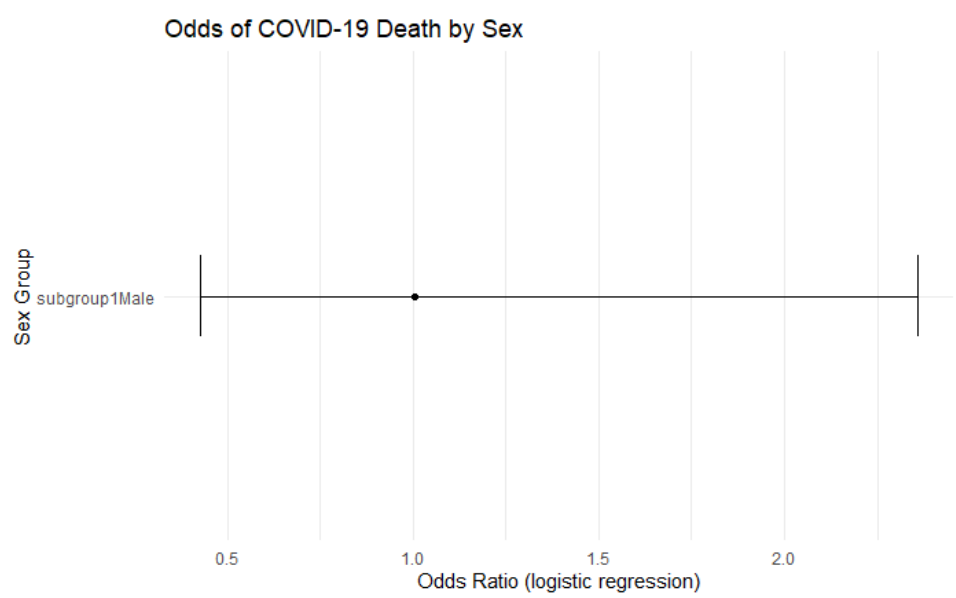
C3: QQ Plot of Race Model



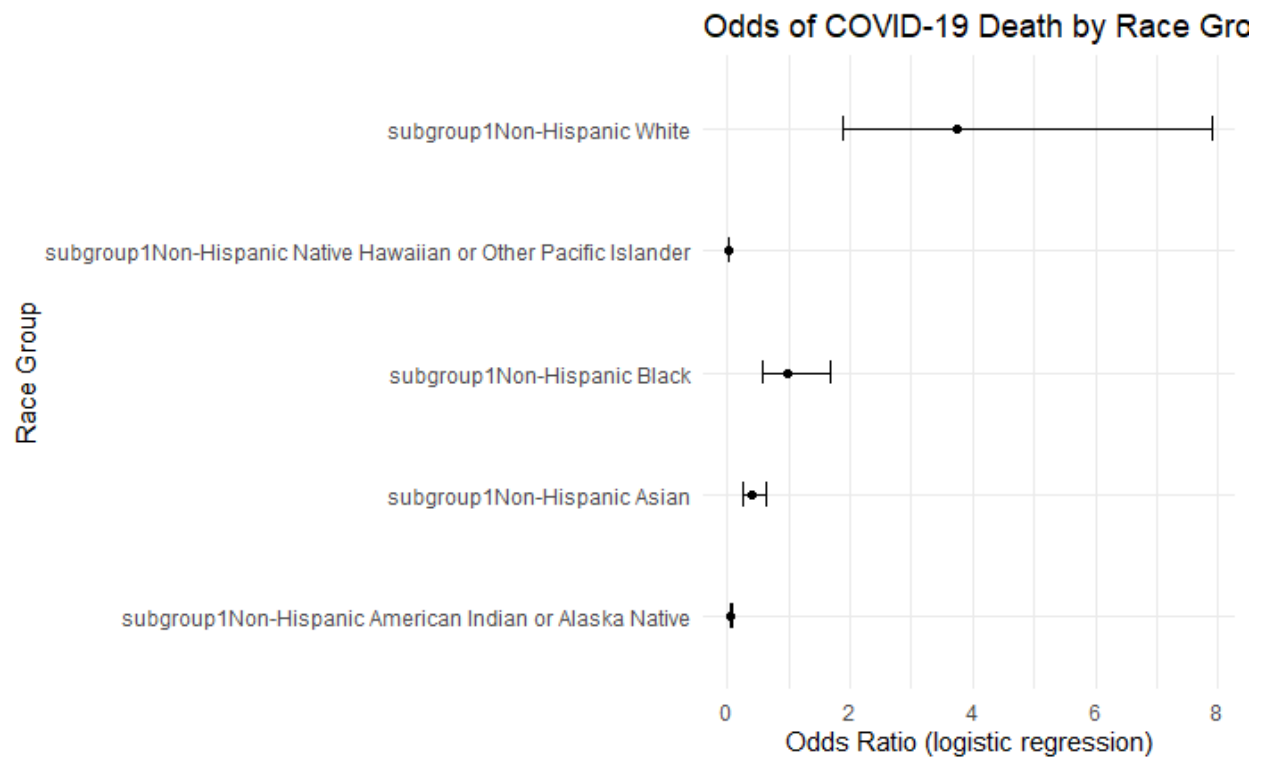
D1:Odds of COVID-19 Death by Age Group



D2:Odds of COVID-19 Death by Sex Group



D3:Odds of COVID-19 Death by Race Group



Appendix B: Full Code + Data

Data Link:

```
```{r setup, include=FALSE}

knitr::opts_chunk$set(echo = TRUE)

```
```

```
# Data Preprocessing
```

Changed the missing data entries from subgroup2 to None since they were reported as missing if not double stratified. Due to the recommendation of the note tab, observations that included <20 deaths but greater than 0 deaths were dropped. This is due to their sensitivity and unreliability. The note column was then dropped.

```
```{r}

library(dplyr)

library(readr)

library(agricolae)

df <- read.csv("D:/UCFFL25/STA4173/CovidDeathRateData.csv")

head(df)

df_clean <- df

df_clean$subgroup2 <- ifelse(df_clean$subgroup2 == "", "None",
df_clean$subgroup2)

df_clean$subgroup2 <- factor(df_clean$subgroup2)

table(df_clean$subgroup2, useNA = "ifany")
```

```

any(is.na(df_clean))

colSums(is.na(df_clean))

df_clean <- df_clean[df_clean$note == "",]

df_clean$note <- NULL

df_clean <- df_clean[complete.cases(df_clean),]
colSums(is.na(df_clean))

summary(df_clean$crude_rate)
...

Summary stats for crude death rates

```{r}
hist(df_clean$crude_rate,
      main = "Histogram of Crude Death Rates",
      xlab = "Crude Death Rate",
      col = "blue")

hist(log(df_clean$crude_rate + 1),
      main = "Histogram of Log Crude Death Rates",
      xlab = "Log Crude Death Rate",
      col = "red")
...

# Summary stats and One-Way ANOVA for Sex

```

```

```{r Sex}

Subset Sex group data
sex_data <- df_clean[df_clean$group == "Sex",]

tapply(sex_data$crude_rate, sex_data$subgroup1, mean) # mean
tapply(sex_data$crude_rate, sex_data$subgroup1, median) # median
tapply(sex_data$crude_rate, sex_data$subgroup1, sd) # standard deviation
tapply(sex_data$crude_rate, sex_data$subgroup1, min) # min
tapply(sex_data$crude_rate, sex_data$subgroup1, max) # max

boxplot(crude_rate ~ subgroup1, data = sex_data,
 main = "COVID-19 Death Rates by Sex",
 xlab = "Sex Group",
 ylab = "Death Rate per 100,000",
 las = 2, # rotate x-axis labels
 col = "lightgreen",
 ylim = c(0,20))

anova_sex <- aov(crude_rate ~ subgroup1, data = sex_data)
summary(anova_sex)

tukey_sex <- TukeyHSD(anova_sex)
tukey_df <- as.data.frame(tukey_sex$subgroup1)
tukey_df_sig <- tukey_df[tukey_df$p_adj` < 0.05,]
tukey_df_sig

qqnorm(residuals(anova_sex), main = "QQ Plot for Sex")
qqline(residuals(anova_sex), col = "lightgreen")

```

```
```
```

```
# Summary stats and One-Way ANOVA for Age
```

```
```{r Age ANOVA}
```

```
Subset Age group data
```

```
age_data <- df_clean[df_clean$group == "Age",]
```

```
#MEANS
```

```
age_means <- tapply(age_data$crude_rate, age_data$subgroup1, mean)
```

```
age_means <- round(age_means, 5)
```

```
age_means_sorted <- sort(age_means)
```

```
age_means_sorted
```

```
#SD
```

```
age_sd <- tapply(age_data$crude_rate, age_data$subgroup1, sd)
```

```
age_sd <- round(age_sd, 5)
```

```
age_sd_sorted <- sort(age_sd)
```

```
age_sd_sorted
```

```
median(age_data$crude_rate)
```

```
sd(age_data$crude_rate)
```

```
tapply(age_data$crude_rate, age_data$subgroup1, median) # median
```

```
tapply(age_data$crude_rate, age_data$subgroup1, sd) # standard deviation
```

```
tapply(age_data$crude_rate, age_data$subgroup1, min) # min
```

```
tapply(age_data$crude_rate, age_data$subgroup1, max) # max
```

```
boxplot(crude_rate ~ subgroup1, data = age_data,
 main = "COVID-19 Death Rates by Age Group",
 xlab = "Age Group",
 ylab = "Death Rate per 100,000",
 las = 2, # rotate x-axis labels
 col = "blue",
 ylim = c(0,20))
```

```
anova_age <- aov(crude_rate ~ subgroup1, data = age_data)
summary(anova_age)
```

```
tukey_age <- TukeyHSD(anova_age)
tukey_df <- as.data.frame(tukey_age$subgroup1)
tukey_df_sig <- tukey_df[tukey_df`p adj` < 0.05,]
tukey_df_sig
```

```
qqnorm(residuals(anova_age), main = "QQ Plot for Age")
qqline(residuals(anova_age), col = "blue")
```

```
tukey_age_cld <- HSD.test(anova_age, "subgroup1", group = TRUE)
cld_age <- tukey_age_cld$groups
cld_age
...
```

```
Summary stats and One-Way ANOVA for Race
```

```

```{r Race}

# Subset Race group data
race_data <- df_clean[df_clean$group == "Race", ]

race_means <- tapply(race_data$crude_rate, race_data$subgroup1, mean)
race_means <- round(race_means, 5)
race_means_sorted <- sort(race_means)
race_means_sorted

race_sd <- tapply(race_data$crude_rate, race_data$subgroup1, sd)
race_sd <- round(race_sd, 5)
race_sd_sorted <- sort(race_sd)
race_sd_sorted

tapply(race_data$crude_rate, race_data$subgroup1, mean)      # mean
tapply(race_data$crude_rate, race_data$subgroup1, median)    # median
tapply(race_data$crude_rate, race_data$subgroup1, sd)        # standard
deviation
tapply(race_data$crude_rate, race_data$subgroup1, min)       # min
tapply(race_data$crude_rate, race_data$subgroup1, max)       # max

boxplot(crude_rate ~ subgroup1, data = race_data,
        main = "COVID-19 Death Rates by Race Group",
        xlab = "Race Group",
        ylab = "Death Rate per 100,000",
        las = 2, # rotate x-axis labels
        col = "red",
        ylim=c(0,50))

```

```

anova_race <- aov(crude_rate ~ subgroup1, data = race_data)
summary(anova_race)
tukey_race <- TukeyHSD(anova_race)
tukey_df <- as.data.frame(tukey_race$subgroup1)
tukey_df_sig <- tukey_df[tukey_df$`p adj` < 0.05, ]
tukey_df_sig

```

```

qqnorm(residuals(anova_race), main = "QQ Plot for Race")
qqline(residuals(anova_race), col = "red")
plot(TukeyHSD(anova_race))

```

```

tukey_race_cld <- HSD.test(anova_race, "subgroup1", group = TRUE)
cld_race <- tukey_race_cld$groups
cld_race

```

```

...

```

```

# Data Preparation for Logistic Regression

```

```

```{r Logistic Regression Death Present}
df_clean_bin <- df_clean
df_clean_bin$death_present <- ifelse(df_clean_bin$covid_deaths > 0, 1, 0)
...

```

```

Logistic Regression for Sex

```

```
```{r Logistic Regression Sex Death Present}

sex_data_bin <- df_clean_bin[df_clean_bin$group == "Sex", ]

logmodel_sex <- glm(death_present ~ subgroup1, data = sex_data_bin, family =
binomial)

summary(logmodel_sex)

```
```

```
Logistic Regression for Age
```

```
```{r Logistic Regression Age Death Present}

age_data_bin <- df_clean_bin[df_clean_bin$group == "Age", ]

logmodel_age <- glm(death_present ~ subgroup1, data = age_data_bin, family =
binomial)

summary(logmodel_age)

```
```

```
Logistic Regression for Race
```

```
```{r Logistic Regression Race Death Present}

race_data_bin <- df_clean_bin[df_clean_bin$group == "Race", ]

logmodel_race <- glm(death_present ~ subgroup1, data = race_data_bin, family =
binomial)

summary(logmodel_race)

```
```

```
Coefficients
```

```
```{r}
```

```
exp(coef(logmodel_age))
```

```
exp(coef(logmodel_sex))
```

```
exp(coef(logmodel_race))
```

```
```
```

```
Confidence Intervals
```

```
```{r}
```

```
exp(confint(logmodel_age))
```

```
exp(confint(logmodel_sex))
```

```
exp(confint(logmodel_race))
```

```
```
```

```
Kruskal and Wilcox
```

Since the normality assumption was not verified for ANOVA, a Kruskal test was done comparing the medians and then a post-hoc pairwise Wilcox test was done. Although outside of the scope of this course, this was a non-parametric option for our zero-inflated and skewed data.

```
```{r Extra Stuff}
```

```
kruskal.test(crude_rate ~ subgroup1, data = age_data)
```

```
pairwise.wilcox.test(age_data$crude_rate,
```

```
                    age_data$subgroup1,
```

```
                    p.adjust.method = "bonferroni")
```

```
kruskal.test(crude_rate ~ subgroup1, data = sex_data)
```

```
pairwise.wilcox.test(sex_data$crude_rate,
```

```
sex_data$subgroup1,  
p.adjust.method = "bonferroni")  
  
kruskal.test(crude_rate ~ subgroup1, data = race_data)  
pairwise.wilcox.test(race_data$crude_rate,  
                      race_data$subgroup1,  
                      p.adjust.method = "bonferroni")  
...
```

Project Member Effort

We, the project teams members, certify that below is an accurate account of the percentage of effort contributed by each team member in the project and Report.

Project Team Member	Effort
Spencer Leighty	33%
Ethan Gau	33%
Josh C	33%